Arab Academy for Science and Technology Maritime Transport

College of Computing and Information Technology

Computer Science

Introduction To Network

Client/Server Calculator Application Using TCP

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Client/Server Calculator Application Using TCP

1. Introduction

This project aims to explain and implement the **Client/Server Architecture** using the **Transmission Control Protocol (TCP)**. TCP is a reliable, connection-oriented protocol that ensures accurate and complete data delivery between the client and the server. The project focuses on understanding the communication flow between both sides rather than on complex application logic

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2. Project Overview

The project is a calculator application in which the client sends a mathematical expression to the server. The server processes the request and returns the result. This process simulates real-world client/server systems such as web applications and cloud services.

2.1 System Architecture

The system consists of the following main components:

Client Application
Provides a graphical user interface (GUI) for user input.

Server Application
Listens for incoming client connections, processes calculation requests, and sends results back to the client.

Communication between the client and server is established using **TCP sockets**.

2.2 Client/Server Scenario

The workflow of the system can be summarized as follows:

The server starts and waits for incoming client connections.

The client connects to the server using a TCP socket.

The client sends a mathematical expression.

The server calculates the result.

The server sends the result back to the client.

The client displays the result in the GUI.

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3. Implementation

3.1 Server Image

A screenshot showing the server application after startup and while listening for incoming requests.

Figure 1: Screenshot of the server application in listening mode.

3.2 Client Image

A screenshot of the client graphical user interface displaying the calculation result.

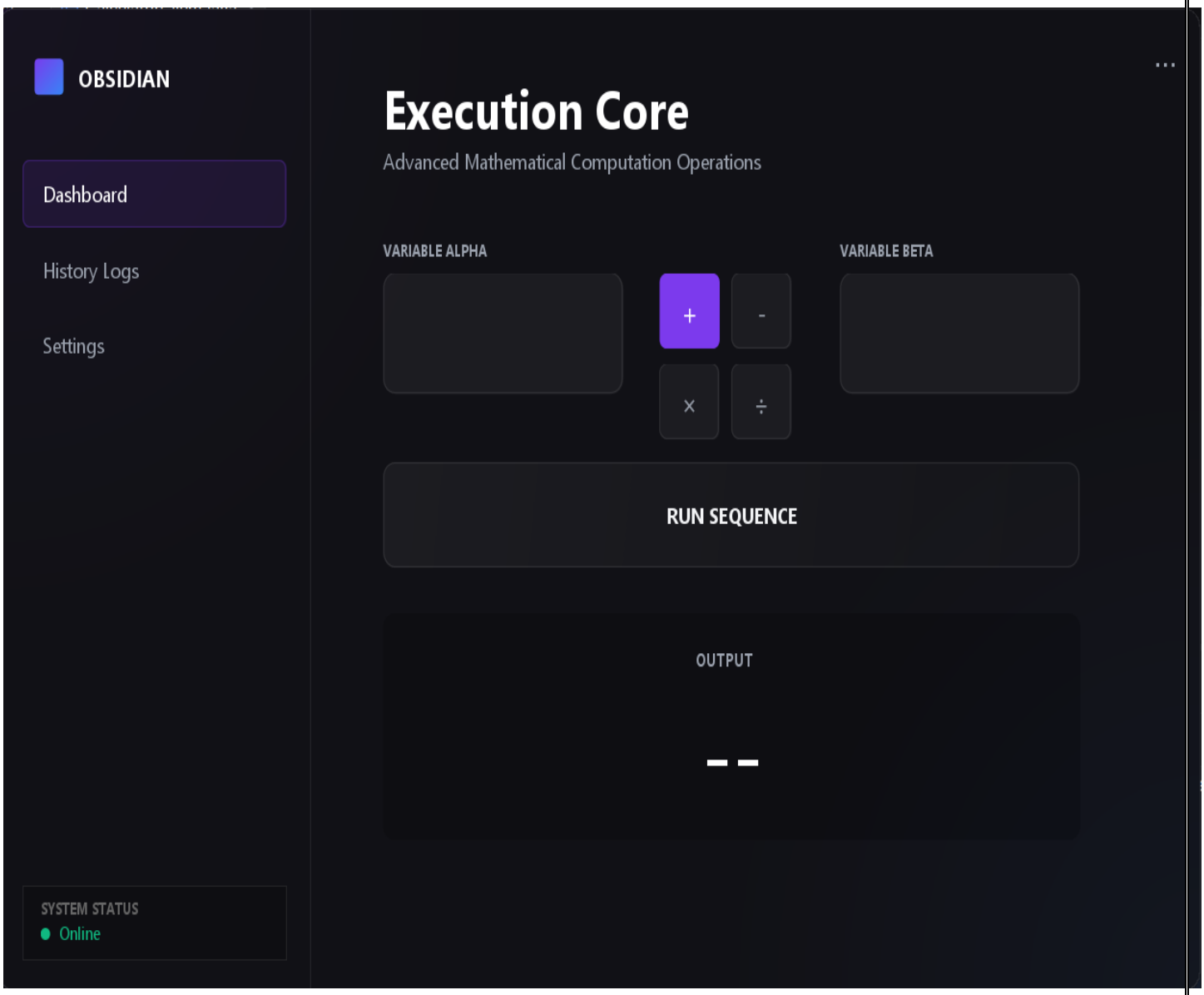
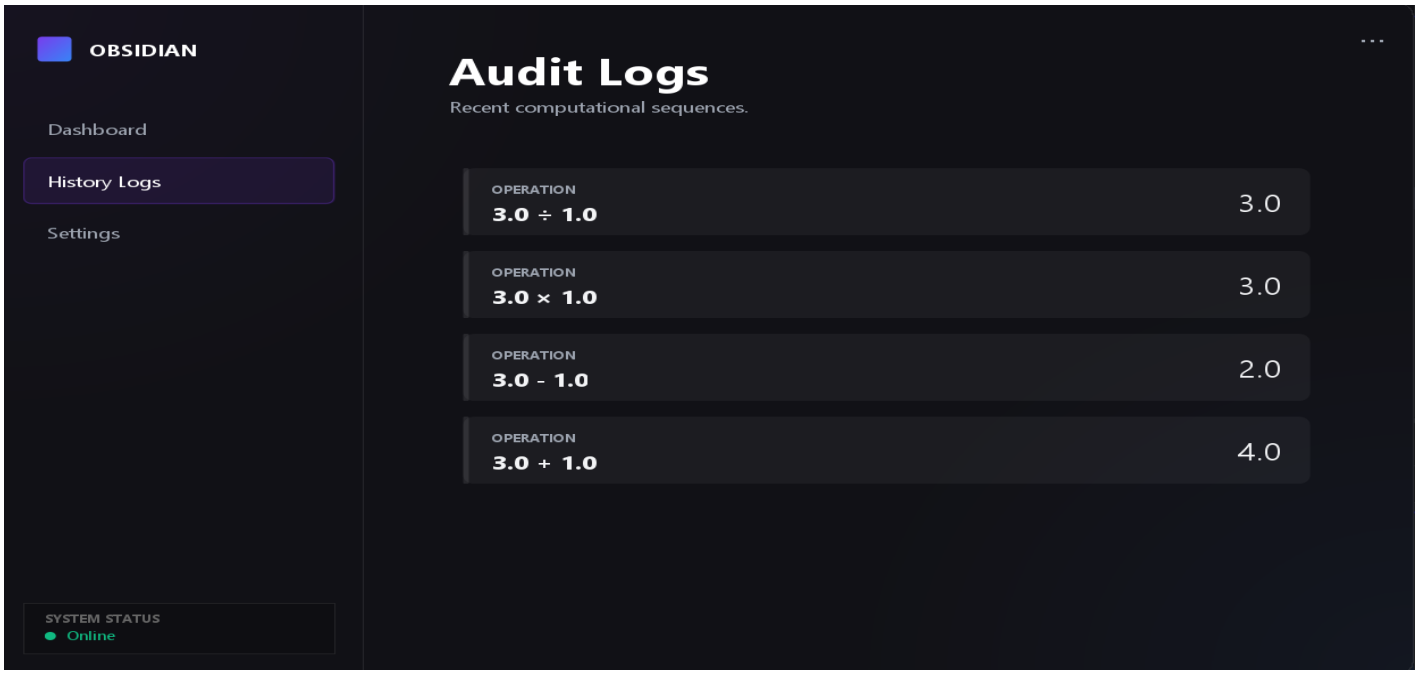
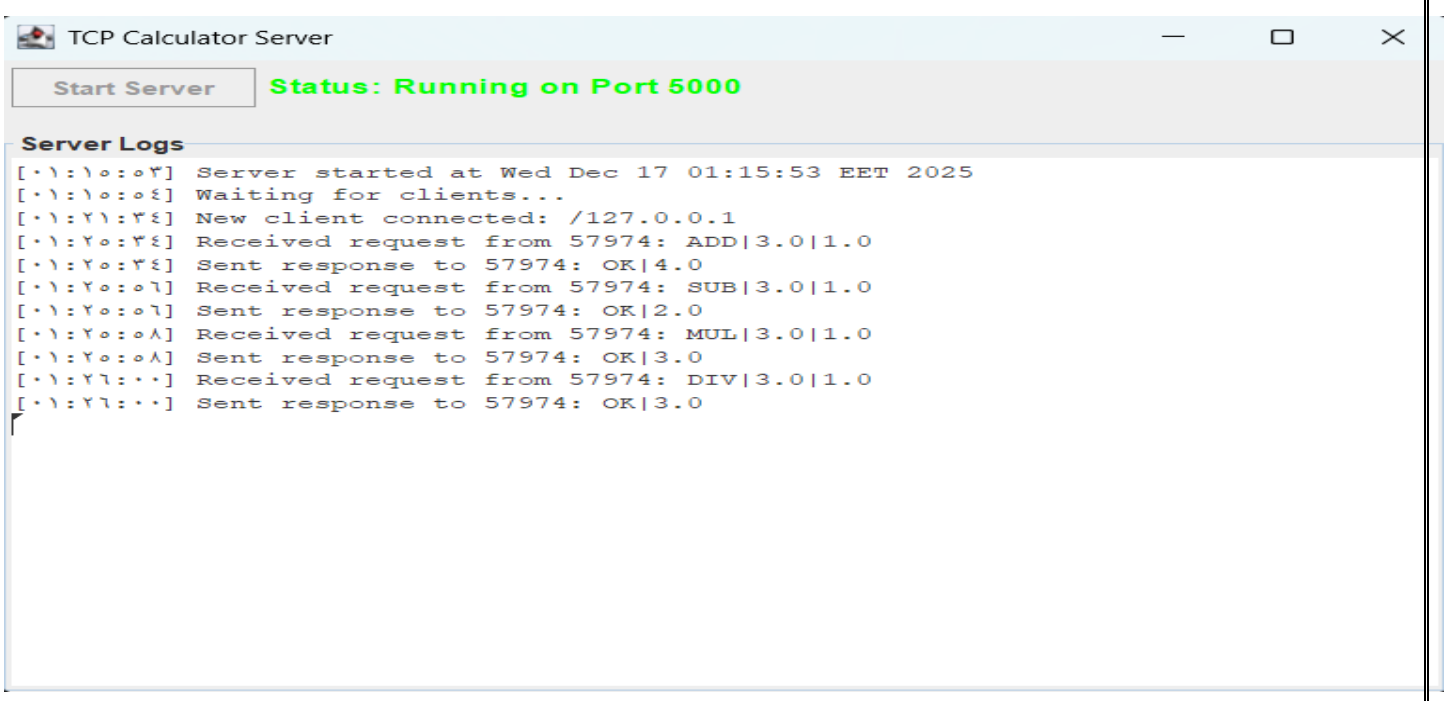


Figure 2: Client GUI calculator before and after sending a calculation request.

3.3 GUI & Out Puts

A simple graphical calculator interface was used to ensure ease of interaction for the user. The results are displayed clearly, allowing users to verify the correctness of the calculations.





3.4 Tools and Technologies

The following tools and technologies were used to implement the project:

Programming Language: Java

Protocol: TCP

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4. Testing and Results

The application was tested using multiple arithmetic operations such as addition, subtraction, multiplication, and division.

All requests were processed correctly by the server.

Reliable responses containing accurate results were received and displayed by the client.

These results confirm the success of the connection-oriented communication and demonstrate the reliability of the TCP protocol in data transmission.

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5. Conclusion

This project helped in understanding the mechanisms of TCP-based client/server communication in a practical way. It demonstrated how data flows between the client and server and how TCP ensures reliable, ordered data delivery, making it suitable for applications that require guaranteed data transmission.

6. References

Computer Networks Lecture Notes

- Java Socket Programming Documentation